



Aerospace Academy Version

Unit 0

Course Launch, Procedures & FabLab Foundations

Unit Overview

Students learn the classroom systems, professional expectations, documentation routines, digital workflow, safety culture, and FabLab access procedures needed before beginning major engineering design work.

Essential Question

How do engineers work safely, professionally, and consistently in a design environment?

Name: _____ Period: _____ Date: _____

Learning Targets

1. Explain the purpose and structure of Introduction to Engineering Design.
2. Follow daily classroom routines and professional engineering expectations.
3. Set up and maintain an engineering notebook.
4. Organize, name, save, and submit digital files correctly.
5. Identify FabLab tools, zones, PPE, safety expectations, and certification requirements.
6. Work with a team to plan, build, test, document, and reflect on a mini engineering challenge.

Unit Deliverables

- Course introduction reflection
- Course readiness checklist
- Engineering notebook setup
- Digital workflow practice
- FabLab safety agreement and safety quiz
- Tool and zone identification sheet
- Mini Engineering Challenge documentation
- Final reflection

Vocabulary

Term	Definition
Engineering Design	
Aerospace Engineering	
FabLab	
Engineering Notebook	
PPE	
Tool Certification	
Workspace Reset	
Professionalism	
Documentation	

Course Readiness Checklist

Item	Complete?
I know where daily instructions are posted	☐
I know how bellwork is completed	☐
I know how work is submitted	☐
I know how to ask for help	☐
I know what to do when absent	☐
I know cleanup expectations	☐
I know tools require permission and certification	☐
I understand unsafe behavior can remove FabLab privileges	☐

Engineering Notebook Setup

Item	Complete?
Name and class period on front page	☐
Course title on front page	☐
Table of contents created	☐
Pages numbered	☐
Daily entry format recorded	☐
Sketch expectations reviewed	☐

Daily Notebook Entry Format

Each entry should include the date, title, objective, notes/planning, evidence of work, and reflection. Engineering notebooks should show thinking, sketches, data, mistakes, redesigns, and next steps.

Digital Workflow

Folder and File Naming

Main folder: IED_LastName_FirstName
Assignment file format: Period_LastName_AssignmentName_Version
CAD file format: Period_LastName_ProjectName_PartName_Version

FabLab Safety Agreement

I agree to use tools only with teacher permission and required certification, wear PPE when required, avoid horseplay, report injuries or damaged tools immediately, use approved materials, clean my workspace, and stop if something seems unsafe.

Student Signature: _____

Date: _____

Safety Quiz Practice

1. When may a student use a FabLab tool?

2. What should you do if a tool appears damaged?

3. Why is cleanup part of safety?

4. What does certification prove?

Mini Engineering Challenge: Mission Control Tower

Scenario

A remote launch site needs a lightweight temporary communication tower. Your team must design a freestanding tower that can support a small antenna payload.

Criteria and Constraints

Criteria	Constraints
Be as tall as possible	Use only provided materials
Stand freely	Fit within the time limit
Support payload for 10 seconds	Be cleaned up completely
Be documented clearly	Follow all safety expectations

Team Roles

Role	Student	Responsibility
Project Lead		Manages time and focus
Documentation Lead		Records sketches and results
Materials Lead		Manages materials and cleanup
Test Lead		Measures and records performance

Initial Sketches

Sketch Space

Sketch Space

Test Results

Test Item	Result
Final height	
Stood freely?	
Held payload for 10 seconds?	
Followed constraints?	

Reflection

What worked?

What failed or changed?

What would you redesign?

Unit 0 Student Work Templates

Engineering Notebook Entry Template

Purpose: Documentation practice

Directions: Use this format for daily notebook entries. Keep enough detail that another person can understand your thinking.

Date: _____

Page: _____

Entry title: _____

Unit: _____

Objective: _____

Notes / planning / evidence:

Re-
flection / next step: _____

Mission Control Tower Planning Grid

Purpose: Isometric sketching

Directions: Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

Part name: _____

Scale: _____

Purpose: _____

Date: _____



Notes / annotations: _____

Mini Engineering Challenge Test Results

Purpose: Testing and analysis

Directions: Record measurements clearly. Include units, observations, failure points, and redesign decisions based on evidence.

Trial	Measurement / Result	Observation	Change Needed
1			
2			
3			
4			
5			

Analysis / redesign notes: